High reheating temperature without axion domain walls
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High reheating temperature without axion domain walls

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Strong CP and QCD axion

- QCD has $\bar{\theta} G \tilde{G}$ (violates CP). However, $\bar{\theta} < 10^{-10}$ (by neutron EDM).
- Anomalous $U(1)_{PQ}$ promotes $\bar{\theta} \to a(x)$, the axion field.
- Strong dynamics makes $\langle a(x) \rangle \to 0$.
- Vafa-Witten theorem also ensures $\bar{\theta} \to 0$ if it is dynamical.

The QCD axion

Many axion models, KSVZ, DFSZ, etc.
Phenomenologically,

$$\mathcal{L} \supset \frac{1}{2}(\partial a)^2 + \frac{1}{32\pi^2} N_{\mathrm{DW}} \frac{a}{f_a} G\tilde{G} + \frac{\partial a}{f_a} \cdot (\cdots) + \underbrace{\delta V}_{\text{explicit breaking}}$$

$$\to \frac{1}{2}(\partial a)^2 + \underbrace{\frac{1}{2} m_a^2 a}_{\text{Non-perturbative}} + \frac{\partial a}{f_a} \cdot (\cdots)$$

- N_{DW} is the domain wall number.
- f_a is usually called decay constant.
- δV causes the axion quality problem.
- The axion mass is given by

$$m_a \simeq 6\,\mu\mathrm{eV} \times \left(\frac{10^{12}\,\mathrm{GeV}}{f_a}\right)$$

The QCD axion

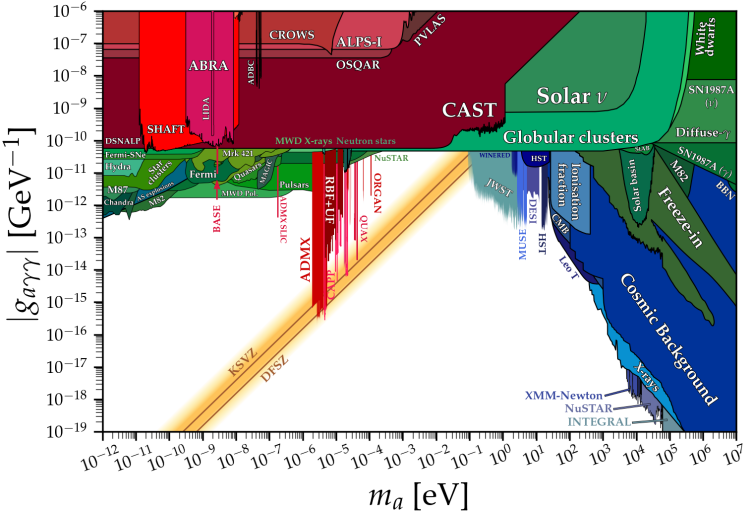


Figure: From AxionLimits.

The axion potential

The axion mass is obtained by expanding the potential around its minimum.
If the quality is ensured, the potential is in the form

$$V(a) = \Lambda^4 \left[1 - \cos\left(\frac{N_{\mathrm{DW}} a}{f_a} \right) \right] .$$

The decay constant f_a, defines the periodicity of the axion field under the shift symmetry.

$$a \to a + \alpha f_a \,, \qquad a + 2\pi f_a \sim a$$

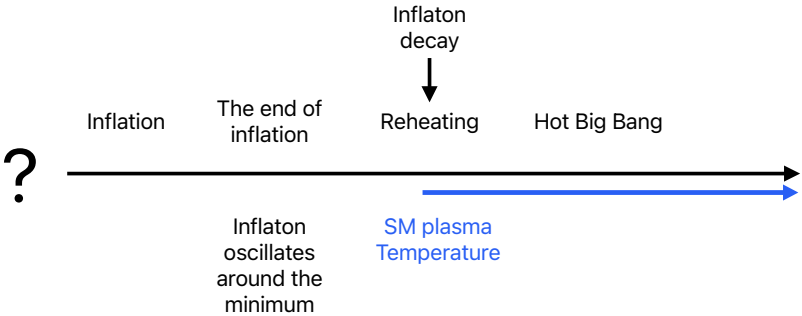
The instanton effect Λ breaks the shift symmetry.
So, N_{DW} determines the number of degenerate vacua.

Axion and PQ field

Shift symmetry is originated from a $U(1)_{\mathrm{PQ}}$. Peccei and Quinn (1977)
The PQ symmetry is spontaneously broken by a PQ scalar $\langle\phi\rangle = v_{\mathrm{PQ}}$.
The decay constant f_a is determined by PQ scale,

$$f_a \propto v_{\mathrm{PQ}}$$

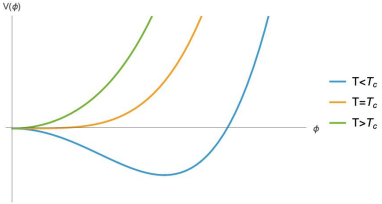
Cosmology



T_R, reheating temperature, describes the hot big bang starting temperature.

Post-inflation

Consider $f_a < T_R$:

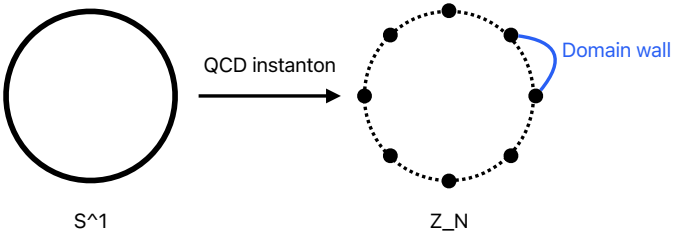


When SM plasma formed, thermal effects will restore the PQ symmetry. Just like EW symmetry.

Here $T_c \sim f_a$.

Defect formation

The universe expands, T becomes smaller. PQ symmetry broken again.



Domain wall forms. $\rightarrow$ domain wall problem

Pre-inflation

Consider $f_a > T_{\rm R}$:
Inflation leads to $a(x) \to a_i$ after inflation.
The PQ symmetry is never restored by SM temperature.
Fluctuations from inflation remain, $\langle \delta a^2 \rangle \simeq H_{\rm inf}/2\pi$.
So one has

$$\theta_{i,\rm eff}^2 = (a/f_a)^2 + (H_{\rm inf}/2\pi f_{\rm inf})^2$$

The CMB power spectrum constrain such fluctuations by

$$\mathcal{P}_{S_c} = \left(\frac{\delta\Omega_a}{\Omega_c}\right)^2 \simeq 1.4 \times \theta_{i,\rm eff}^2 \left(\frac{H_{\rm inf}}{2\pi f_{\rm inf}}\right)^2 \left(\frac{f_a}{10^{12}\,\rm GeV}\right)^{2.38} < 10^{-11}$$

For $f_a = 10^{12}$ GeV, one has $H_{\rm inf} < 10^8$ GeV. Isocurvature problem

Any way out?

High reheating temperature is good for theories like leptogenesis.
Usually, T_{R} is limited by the H_{inf} from above.
How can we have High reheating temperature without domain wall problem and isocurvature problems?

Any way out?

A symmetry nonrestoration mechanism!

The thermal potential

Consider a scalar ϕ in the high temperature limit.
Suppose we have another scalar s couple to it through $\lambda_{\phi s}\phi^2 s^2$, we have

$$V^\beta \supset \lambda_{\phi s}\phi^2 T^2 + \cdots$$

So, if $\lambda_{\phi s} < 0$, it protects the broken phase.

Back to isocurvature

If we succeed in protects the broken phase against the T_{R},
the axion is just in pre-inflation scenario, which has isocurvature problem.

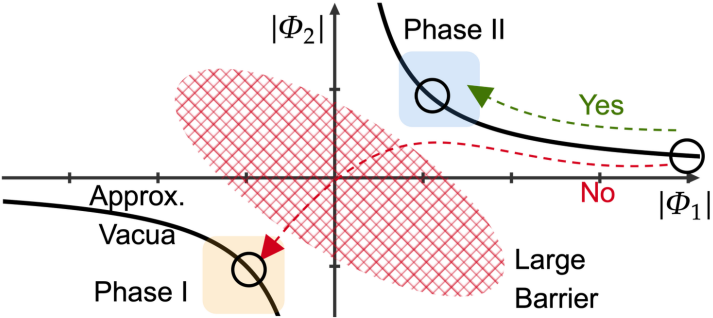
Not if $f_{\mathrm{inf}} \gg f_a$ (Linde's (1991)).

Recall,

$$\mathcal{P}_{S_c} = \left(\frac{\delta\Omega_a}{\Omega_c}\right)^2 \simeq 1.4 \times \theta_{i,\mathrm{eff}}^2 \left(\frac{H_{\mathrm{inf}}}{2\pi f_{\mathrm{inf}}}\right)^2 \left(\frac{f_a}{10^{12}\,\mathrm{GeV}}\right)^{2.38} < 10^{-11}$$

The parametric production problem

Schematic realization of avoid parametric production
Kasuya, Kawasaki, Yanagida (1997),
Kawasaki, Sonomoto (2018),
Kawasaki, Sonomoto, Yanagida (2018)



The bird-eye view

During inflation, the PQ fields stabilize at large field values, leads to $f_{\text{inf}} \gg f_a$.
After inflation, the PQ fields starts to rolling towards its global minimum, which eventually gives f_a.
The existence of the flat direction suppresses the parametric production of axion.
When reheating completed, thermal correction will not restore the PQ symmetry thanks to the additional scalar s.

The model

Suppose that axion is originated from two complex scalar field, $\Phi_+(+1)$ and $\Phi_-(-1)$.

$$V_0 = \lambda \left|\Phi_+\Phi_- - v^2\right|^2 + m_+^2 \left|\Phi_+\right|^2 + m_-^2 \left|\Phi_-\right|^2 + \lambda_{\phi s} s^2 \left(\Phi_+\Phi_- + \text{c.c.}\right)$$

In the limit of $m_\pm \to 0$ and $\lambda_{\phi s} \to 0$, flat direction is $\Phi_+\Phi_- = v^2$.

$$\Phi_+ = \frac{\Phi_1 + \Phi_2}{\sqrt{2}}\,, \quad \Phi_- = \frac{\Phi_1^* - \Phi_2^*}{\sqrt{2}}$$

$$\Phi_1 = \frac{\phi}{\sqrt{2}} e^{ia/v_{\rm PQ}}\,, \quad \Phi_2 = \frac{\xi + i\eta}{\sqrt{2}} e^{ia/v_{\rm PQ}}\,,$$

a is the axion and $\{\phi, \eta, \xi\}$ are three PQ scalars, $v_{\rm PQ} = \sqrt{\langle\phi\rangle^2 + \langle\xi\rangle^2 + \langle\eta\rangle^2}$.

The model

$$
\begin{aligned}
V_0 = & -\frac{1}{2}\mu_\phi^2\phi^2 + \frac{1}{2}\mu_\xi^2\xi^2 + \frac{1}{2}\mu_\eta^2\eta^2 \\
& + \frac{\lambda}{8}\left(\eta^2\phi^2 - \phi^2\xi^2 + \eta^2\xi^2\right) + \frac{\lambda}{16}\left(\phi^4 + \xi^4 + \eta^4\right) \\
& + \frac{\lambda_{\phi s}}{2}s^2\phi^2 - \frac{\lambda_{\phi s}}{2}s^2\left(\xi^2 + \eta^2\right) \\
& + y_j\phi\bar{\psi}_j\psi_j + \frac{\mu_s^2}{2}s^2 + \frac{\lambda_s}{4}s^4 \,,
\end{aligned}
$$

The thermal potential

$$V_{\rm eff}(\phi,\xi,\eta,s,T) = V_0 + V_1 + \cdots \ ,$$

$$V_1 = \sum_{i=\phi,\xi,\eta,s,\psi_j} \frac{n_i T}{2} \sum_{n=-\infty}^{\infty} \int \frac{d^3\vec{k}}{(2\pi)^3} \log\left[\vec{k}^2 + \omega_n^2 + m_i^2(\phi,\xi,\eta,s) + \Pi_i(T)\right] \ ,$$

$$m_\phi^2 = -\mu_\phi^2 + \lambda_{\phi s} s^2 + \frac{3\lambda}{4}\phi^2 - \frac{\lambda}{4}\xi^2 + \frac{\lambda}{4}\eta^2 \ , \qquad \Pi_\phi = \left(\sum_j \frac{y_j^2}{2} + \frac{\lambda}{16} + N_s \frac{\lambda_{\phi s}}{12}\right) T^2 \ ,$$

$$m_\xi^2 = \mu_\xi^2 - \lambda_{\phi s} s^2 - \frac{\lambda}{4}\phi^2 + \frac{3\lambda}{4}\xi^2 + \frac{\lambda}{4}\eta^2 \ , \qquad \Pi_\xi = \left(\frac{\lambda}{16} - N_s \frac{\lambda_{\phi s}}{12}\right) T^2 \ ,$$

$$m_\eta^2 = \mu_\eta^2 - \lambda_{\phi s} s^2 + \frac{\lambda}{4}\phi^2 + \frac{\lambda}{4}\xi^2 + \frac{3\lambda}{4}\eta^2 \ , \qquad \Pi_\eta = \left(\frac{5\lambda}{48} - N_s \frac{\lambda_{\phi s}}{12}\right) T^2 \ ,$$

$$m_s^2 = \mu_s^2 + 3\lambda_s s^2 + \lambda_{\phi s}\phi^2 - \lambda_{\phi s}\left(\xi^2 + \eta^2\right) \ , \qquad \Pi_s = \left((N_s+2)\frac{\lambda_s}{12} - \frac{\lambda_{\phi s}}{12}\right) T^2 \ ,$$

$$m_{\psi_j}^2 = y_j^2 \phi^2 \ .$$

The thermal potential

$$V_1 = V_1^{(0)} + V_1^{\beta} + V_{\text{daisy}}$$

$$V_1^{(0)} = \sum_{i=\phi,\xi,\eta,s,\psi_j} \frac{n_i m_i^4}{64\pi^2}\left(\log\frac{m_i^2}{m_{0i}^2} - \frac{3}{2}\right) ,$$

$$V_1^{\beta} = \sum_{i=\phi,\xi,\eta,s} \frac{n_i T^4}{2\pi^2} J_b\left(\frac{m_i^2}{T^2}\right) + \sum_{j=1}^{N_\psi} \frac{n_{\psi_j} T^4}{2\pi^2} J_f\left(\frac{m_i^2}{T^2}\right) ,$$

$$V_{\text{daisy}} = \sum_{i=\phi,\xi,\eta,s} \frac{n_i T}{12\pi}\left[m_i^3 - \left(m_i^2 + \Pi_i\right)^{3/2}\right] ,$$

The parameter space

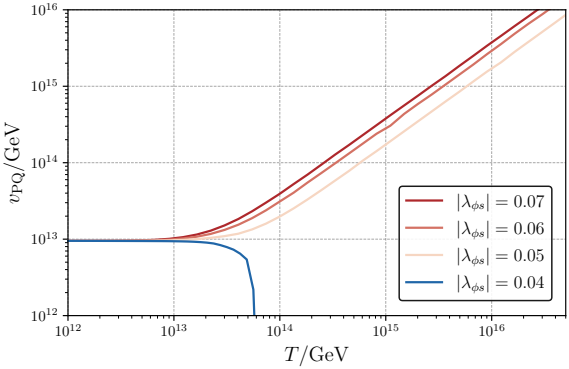


Figure: The PQ scale $v_{\rm PQ}$ against temperature T. Here we take $f_a = 10^{12}$ GeV, $\lambda = 0.03$, $m_\psi = 10^{11}$ GeV, $\mu_s = 4 \times 10^{12}$ GeV, $\mu_\xi = \mu_\eta = 2 \times 10^{12}$ GeV, $\lambda_s = 0.8$, $N_s = 1$ and $N_{\rm DW} = N_\psi = 10$.

The parameter space

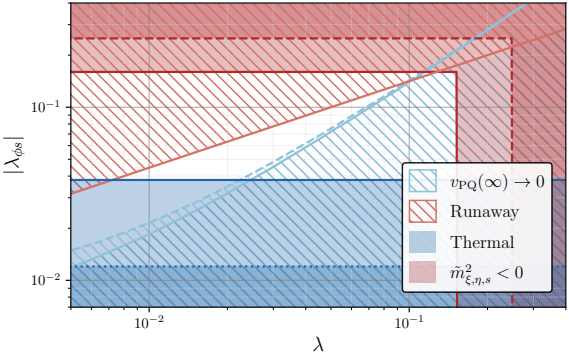


Figure: The runaway region is where $|\lambda_{\phi s}| > \sqrt{\lambda_s \lambda}/2$. The $v_{\rm PQ}(\infty) \to 0$ indicates a failed protection. The $\tilde{m}^2_{\xi,\eta,s} < 0$ region indicates the instability of expected vacuum solution. The solid (dashed) lines are limits under $f_a = 10^{12}$ (8×10^{11}) GeV. The thermal bounds (blue shaded region) are where s does not have thermal contact with PQ fields; the solid (dotted) line is obtained under $T_{\rm R} = 10^{15}$ (10^{14}) GeV.

Takeaway

- Low reheating temperature gives pre-inflation axion, which usually has isocurvature problem.
- High reheating temperature gives post-inflation axion, which usually has domain wall problem.
- Our model allow high reheating temperature and free from both problems.